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Why Engineering Excellence Gets Lost in Economic Translation

February 2026

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Executive Summary

Discrete manufacturing is facing a structural innovation bottleneck. While industry leaders widely acknowledge that *"80% of environmental impact is determined in the design phase — that's where the lever is,"* organizations are structurally unable to pull that lever effectively. Engineers operate in the physical world of materials, tolerances, and thermodynamics — yet final decisions are made in economic spaces defined by cost, risk, and quarterly targets. In this reality, **the price tag remains the ultimate gatekeeper.**

Our research, based on more than 40 interviews across the industrial design ecosystem, reveals four interconnected barriers to sustainable, profitable innovation.

First, design teams work in a **data desert**. Critical information, i.a., CO_2 footprints, supplier constraints, lifecycle costs, and regulatory requirements, is fragmented across PLM, ERP, spreadsheets, and static documents. The data exists, but not where and when decisions are made. As a result, early-stage design relies heavily on assumptions and "gut feeling" instead of real-time intelligence.

Second, organizations are trapped in a costly **Redesign Trap**. Because cost and manufacturability implications surface too late, teams are forced into expensive rework during prototyping or production. Every late change multiplies cost and delays time-to-market — turning avoidable misalignment into systemic inefficiency.

Third, the industry is experiencing a silent **erosion of tribal knowledge**. Critical experiential know-how — why certain materials fail, which production shortcuts create downstream risks — often resides in the minds of senior engineers. As they retire, this undocumented decision logic disappears. Younger teams are left digitally equipped yet context-poor, repeatedly reinventing solutions rather than building on institutional memory.

Across all challenges, immediate cost dominance shapes behavior. Sustainability is pursued only if cost-neutral, legally required, or marketable. Long-term lifecycle optimization rarely survives short-term financial scrutiny.

To break this cycle, manufacturers must transition toward **Circular Engineering**, supported by an intelligent digital layer. **AI-powered copilots embedded directly into design environments** can provide real-time feedback on cost, carbon impact, compliance, and manufacturability — while simultaneously capturing and reusing institutional knowledge.

Aligning engineering physics with economic logic at the design stage is the decisive shift. When real-time decision intelligence replaces late-stage correction, sustainability and profitability stop competing — and start reinforcing each other.

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1. Introduction

The Problem Space

The global manufacturing sector is undergoing a fundamental transformation in how products are conceived, engineered, and managed across their lifecycles. Electrification, digital integration, and tightening regulations such as the EU's Digital Product Passport (DPP) and Corporate Sustainability Reporting Directive (CSRD) are accelerating development cycles beyond what legacy systems were designed to handle. What was once an incremental improvement has become a structural reinvention.

Yet inside many organizations, a disconnect persists. Corporate boards define ambitious ESG targets and profitability goals, while engineering teams operate in fragmented data environments, relying on manual consolidation, spreadsheets, and experience-based judgment. At the same time, a demographic shift threatens to erase decades of undocumented expertise as senior engineers retire.

This convergence of regulatory pressure, economic constraint, and knowledge loss has created what we call an industrial complexity crisis.

At the center of this crisis lies the design lever.

The design lever refers to the structural reality that early engineering decisions determine up to 80% of a product's lifecycle cost, environmental footprint, manufacturability, and service complexity. Geometry, material choice, tolerances, architecture, and supplier selection lock in downstream consequences long before production begins. Once these parameters are fixed, change becomes exponentially more expensive.

In other words: **Design is the point of maximum influence and minimum cost of change.**

A Systemic Problem; Not a Sector-Specific One

This white paper is grounded in **44 in-depth qualitative interviews** conducted across the industrial design ecosystem. Our interview partners include CEOs, general managers, R&D heads, senior vice presidents of engineering, product managers, sustainability leaders, design engineers, process engineers, implementation consultants, investors, and academic researchers.

They represent a broad cross-section of discrete manufacturing industries: automotive, aerospace, electronics, packaging, rail rolling stock, battery systems, tooling, manufacturing machinery, material handling, cooling technologies, building components, engineering services, sustainability consulting, and engineering

software providers. The sample spans large industrial players, mid-sized manufacturers, specialized engineering firms, software vendors, and academia.

This diversity reveals that the bottlenecks identified are not isolated to one sector — they are systemic.

Across interviews, a recurring pattern emerged: the tools used in early-stage design are disconnected from real-time cost intelligence, lifecycle data, procurement constraints, and regulatory requirements. Engineers work in physical and chemical realities — stress curves, tolerances, thermodynamics — yet final decisions are governed by economic metrics. In this environment, the "price tag" functions as the ultimate gatekeeper.

Moving toward an **Intelligent Design Lever** — supported by integrated data environments and AI-powered co-pilots embedded directly within engineering tools — fundamentally changes this dynamic:

- **Cost:** Early validation reduces redesign loops, shortens development cycles, improves margin predictability, and prevents late-stage cost surprises.
- **Environment:** Real-time CO_2 and circularity feedback enables measurable lifecycle optimization before lock-in.
- **Organization:** Tribal knowledge is captured digitally, cross-functional alignment improves, and silo-driven friction decreases.

Mapping the Friction Points: From Economic Constraint to Intelligent Design

This white paper maps the structural friction points preventing organizations from activating this lever — from cost dominance and data fragmentation to knowledge erosion and late-stage redesign cycles. The paper is structured to guide the reader through the various "friction points" discovered during our research, moving from immediate economic barriers to the long-term potential for technological solutions:

- **Chapter 2: The Economic Gatekeeper** explores why the "price tag" remains the ultimate decider for every technical selection, often reducing sustainability to a mere "image factor" rather than a driver of core innovation.
- **Chapter 3: The Data Desert** analyzes the systemic fragmentation of product data, detailing how the lack of a "single source of truth" leads to a heavy reliance on manual "Excel-hell" and significant delays in information retrieval.
- **Chapter 4: The Erosion of Tribal Knowledge** highlights the demographic risk facing the industry as a generation of experienced engineers retires, taking decades of unrecorded "tribal knowledge" with them.
- **Chapter 5: The Redesign Trap** details the massive financial impact of late-stage feedback, where a disconnect between design and manufacturing

forces expensive rework only discovered during prototyping or series production.

- **Chapter 6: Sustainability as a Compliance Burden** examines why environmental reporting is currently viewed as a "painful process" and a legal hurdle rather than a business opportunity.
- **Chapter 7: The Intelligent Design Layer** outlines the opportunity for AI-powered "co-pilots" to bridge these gaps, automating manual tasks and providing real-time decision intelligence.

By mapping these obstacles, this report provides a framework for "Circular Engineering," a model that aligns financial profitability with environmental responsibility by pulling the most powerful lever available. By transforming early-stage engineering from assumption-driven to intelligence-driven, manufacturers can align economic performance and environmental responsibility at the moment where it matters most.

The future of competitive manufacturing will not be decided on the shop floor — but at the design desk.

2. The Economic Gatekeeper: Profit vs. Purpose

In the modern manufacturing landscape, sustainability is often heralded in marketing materials, yet the operational reality is governed by a singular, uncompromising metric: immediate cost. Throughout our interviews, the sentiment that economic viability remains the absolute boundary for innovation was near-universal. As one CEO noted during the research, **"no one cares about sustainability unless it means money,"** reflecting a culture where environmental goals are only pursued if they are cost-neutral or mandated by law. This financial pressure creates a hierarchy of priorities where **"the price tag is the one thing that decides all and everything,"** frequently rendering sustainability as a mere "image factor" that carries little weight in the final technical selection.

Project managers and lead engineers describe a persistent struggle to build a financial "business case" for circularity. In many sectors, the market currently refuses to accept premium pricing for sustainable products, leading some firms to remove "green" terminology from their pitches to avoid negative sales reactions. Engineers find themselves in a position where they must prioritize "low-hanging fruit"—choices that simultaneously reduce material usage and carbon footprints—because fundamental architectural shifts toward sustainability are deemed too expensive to pursue. Within the boardroom, the tension remains high as **"the primary question is always 'what does it cost?'... everything else is secondary"** to the immediate EBIT and EBITDA targets.

3. The Data Desert: Fragmentation and Silos

Despite the adoption of sophisticated Product Lifecycle Management (PLM) and Enterprise Resource Planning (ERP) systems, the "single source of truth" remains an industrial myth for most practitioners. Critical product data—including material CO2 footprints, current supplier pricing, and manufacturing constraints—is often scattered across incompatible systems. As one PLM consultant observed, **"product data resides in multiple different enterprise applications... the silos don't work anymore,"** yet the manual effort required to bridge these gaps is immense.

This fragmentation leads to a heavy reliance on "Excel-hell," where teams spend days manually consolidating information to prepare for design freezes or board reviews. Professionals report that **"the tools could today already do much more than we use... we are still using plain Word and Excel"** for high-stakes decision-making. This lack of integration means that retrieving a complete history of a component's usage can take a team several days because the data is trapped in manual assembly drawings or local spreadsheets rather than an integrated database. The inefficiency of this "abstract data mess" prevents designers from performing the real-time "What-If" scenarios necessary to optimize a product's lifecycle.

4. The Erosion of Tribal Knowledge

A critical but often overlooked bottleneck is the manufacturing industry's heavy reliance on undocumented expertise. Significant engineering wisdom—knowing which materials "actually hold up" in extreme environments or which production methods are likely to fail on the shop floor—resides primarily in the minds of a few senior employees. A Team Lead succinctly described this fragility: **"I call it Tribal-Knowledge. That is just with a few people, they know it all"**.

As this demographic reaches retirement age, organizations are facing a massive "knowledge drain". Junior designers, while proficient in modern CAD software, often lack the historical context needed to avoid repeating costly past mistakes. Without digital tools to codify and reuse institutional memory, new teams are forced to "reinvent the wheel" with every project. One expert noted that **"the junior doesn't know it all... the decision whether to go left or right is usually not possible with his background,"** highlighting the urgent need for systems that act as a digital mentor by capturing the reasoning behind past engineering choices.

5. The Redesign Trap: The Cost of Late Feedback

The disconnect between the design office and the factory floor creates a "redesign trap" that bleeds profitability. Decisions regarding geometry and material are often made with incomplete requirements, leading to "surprises" only discovered during the prototyping or manufacturing phase. A Senior Designer expressed the common frustration of this delayed flow: **"Ideally, this was supposed to happen a lot earlier in our design phase... now we have to change this, and it's like, why haven't I got this information earlier?"**.

Because design is often decoupled from procurement, **"designers are often the last to know about the final price... there are sometimes surprises at the last stage"** that force immediate and expensive redesigns. As soon as a project enters the manufacturing process, every modification becomes exponentially more costly, yet these changes are frequently triggered by simple communication lapses. For example, in aerospace, a minor lack of symmetry in 3D models once led to thousands of wasted installation hours because wire harnesses could not be physically installed as designed. This friction is exacerbated when technicians on the shop floor cannot understand the intent of design notes, leading to further rework and delays.

6. Sustainability: From Compliance Burden to Innovation

Currently, sustainability is largely treated as a reporting hurdle rather than a fundamental innovation driver. Professionals describe a "painful process" where environmental metrics are calculated *after* the design is finalized, primarily to satisfy legal requirements like the EU's CSRD. One sustainability manager noted that **"it's a bit of a painful process... the board sends us back to recalculate, and it takes such a long time"**. This indicates that many organizations view changing regulations as an "enemy" to their work rather than a roadmap for competitive advantage.

To move toward "Circular Engineering," sustainability data must be integrated into the tools engineers use every day. Until designers have real-time access to the CO2 and circularity implications of their choices, environmental impact will remain an "afterthought" that is impossible to optimize without compromising performance or cost. The research suggests that the path forward lies in bridging the gap between strategic board-level goals and the localized "data mess" of the actual design office.

7. Opportunity Space: The Intelligent Design Layer

The interviews revealed a strong consensus on the need for an "intelligence layer" to help designers manage modern industrial complexity. This tool would ideally function as an AI-powered co-pilot that lives within or alongside CAD systems, providing real-time feedback on cost, carbon impact, and regulatory compliance during the modeling phase. Such a system would enable engineers to move beyond "gut feeling" toward evidence-based decisions.

By automating the "commodity tasks" of engineering, such as manual documentation and requirement checking, designers can focus on higher-level strategy and quality assurance. This shift allows for the creation of more "agile and flexible" workflows that can respond to shifting market conditions and regulatory changes without triggering the "Redesign Trap". One managing director highlighted the stakes of this shift, noting that **"if I really want to develop something new, it takes four years... and lower seven-digit development costs,"** emphasizing that any tool that reduces this timeframe offers a massive competitive advantage.

8. Recommendations & Conclusion

To navigate the future of discrete manufacturing, organizations must prioritize the following actions:

1. **Break Down Data Silos:** Transition away from disconnected "pots" of data toward a "single source of truth" by unifying PLM, ERP, and CAD systems.
2. **Codify Tribal Knowledge:** Implement systems to document the "why" behind engineering choices, ensuring corporate memory survives the retirement of senior staff.
3. **Involve Cross-Functional Teams Early:** Bring manufacturing, procurement, and service technicians into the conceptual phase to avoid late-stage "surprises".
4. **Prioritize Decision Intelligence:** Invest in tools that provide real-time feedback on cost and carbon impact *during* the design process, rather than as a post-production report.

The industrial design ecosystem is at a crossroads where legacy workflows can no longer meet the demands of a globalized, regulated market. By transforming the design process into a data-driven discipline in which environmental and financial optimizations occur simultaneously, manufacturers can turn the design phase into an engine for sustainable growth.

Appendix: List of Interviewees

I#	Job title of the interviewee	C#	Industry of the company
I1	Implementation Consultant, Co-Founder	C1	Engineering software implementation consulting
I2	Implementation Consultant, Co-Founder	C1	Engineering software implementation consulting
I3	Head of Brand and Design, Founder	C2	Engineering Services
I4	General Manager	C3	Packaging
I5	CEO	C4	Electronics
I6	Business Developer, Co-Founder	C5	Production software provider
I7	Investor	C6	Production software provider
I8	Head of Development and Pre-series Production	C7	Automotive
I9	R&D Head	C8	Building shadings
I10	R&D Head	C9	Electronics
I11	Mechanical Engineer	C10	Engineering Services
I12	Mechanical Engineer	C11	Academia
I13	Co-Founder	C12	Engineering Services
I14	Process Engineer	C13	Packaging
I15	Senior Director Of Strategy	C14	Consulting
I16	Postdoctoral Research	C15	Academia
I17	General Manager	C16	Electronics
I18	Consultant, Founder	C17	Sustainability Consulting
I19	Product Designer, Founder	C18	Engineering Services
I20	Design Engineer	C19	Automotive
I21	Civil Engineer	C20	Engineering Services
I22	Team Leader Strategy Standard Products	C4	Electronics
I23	Product Designer Thermal Solutions	C21	Engineering Services
I24	Product and Process Engineer	C22	Automotive
I25	Product Development Team Lead	C22	Automotive
I26	Project Manager	C23	Electronics
I27	Product & Process Engineer	C22	Automotive
I28	Consultant, Founder	C24	Engineering Services
I29	Advance Design Engineer	C22	Automotive

I30	Consultant	C25	Engineering software implementation consulting
I31	Sustainability Manager	C26	Material handling
I32	Product Manager	C26	Material handling
I33	Product Developer	C27	Rail rolling stock
I34	CIO, Software Engineer	C28	Cooling
I35	Design Engineer	C29	Manufacturing machinery
I36	CEO	C29	Manufacturing machinery
I37	Master of Engineering	C30	Battery
I38	Senior Vice President Design & Engineering	C31	Tooling
I39	Research and Development Engineer	C32	Automotive
I40	Product Designer	C33	Product Design
I41	Design Engineer	C34	Aerospace
I42	Repair Design Engineer	C35	Aerospace
I43	Program Manager Product Sustainability	C26	Material handling
I44	Head of Product Management	C36	Material handling

Note: I#-Interviewee Number, C#-Company Number

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